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Editor-in-Chief *Sensors* (MDPI)

Re: Submission of original research article “Causal-Consolidated SHAP: A Lightweight, Robustness-Oriented Explainer for Resource-Constrained IoT Edge Intrusion Detection”

Dear Editor,

We are pleased to submit the above manuscript for consideration as a Research Article in *Sensors*, and request that it be reviewed in the Sensor Networks section.

Summary of contribution. Explainable AI (XAI) methods applied to IoT edge intrusion detection are routinely evaluated on accuracy or, at best, explanation fidelity alone, with little attention to whether the explanation itself remains affordable, stable, and robust under the resource and adversarial constraints of real edge deployment. This gap was formalized, but not empirically closed, by a recent *Future Internet* survey (Karras et al., 2026), which defines a hardware-centric evaluation framework (Computational Complexity Score, Memory Footprint Ratio, Temporal Coherence, and a composite UMCEF_IoT score) entirely by equation, without computing it on real data. Our manuscript makes two linked contributions:

1. We implement this full evaluation framework and apply it to four established post-hoc explainers (TreeSHAP, KernelSHAP, a lightweight KernelSHAP variant, and LIME) across two edge-representative models (LightGBM and a compact multilayer perceptron) under three simulated resource tiers (Raspberry Pi 4-, Jetson Nano-, and unconstrained-class budgets), at production scale on two structurally different real-world datasets, CICIOT2023 and Edge-IIoTset.
2. We propose CC-SHAP (Causal-Consolidated SHAP), a lightweight post-processing explainer that answers the field’s explicit call for causal, network-native alternatives to generic tabular SHAP/LIME. CC-SHAP is the most adversarially robust and by far the sparsest explainer of the five tested, on both datasets, at a real and transparently reported fidelity cost. A dedicated ablation study further tests, and partially qualifies, the causal mechanism behind this result, a level of self-scrutiny we believe strengthens rather than weakens the contribution.

Fit to the journal. Both datasets used in this study have their most-cited companion papers published in *Sensors* (the CICIOT2023 dataset paper itself, and the DeepFool-based explanation-robustness protocol we replicate and extend), giving this manuscript strong topical alignment with the journal’s existing body of work on IoT/IIoT sensing, security, and edge intelligence.

Originality and prior disclosure. This manuscript is original, has not been published previously, and is not under consideration at any other journal. All authors have approved the manuscript and agree to its submission. The authors declare no conflicts of interest.

Data and code availability. The full experimental pipeline, trained-model configurations, and all result files reported in this manuscript are released as open-source code at <https://github.com/Ajay102456/cc-shap-iot-edge-xai>, consistent with *Sensors*’ data-sharing policy.

Use of AI tools. As disclosed in the manuscript’s Acknowledgments, large language model-based tools were used to assist with code implementation and execution of the experimental simulations under the authors’ direction; the research design, interpretation, and writing are the authors’ own, and all reported results were independently verified by both authors against the underlying code and data prior to submission.

We believe this manuscript will be of interest to *Sensors*’ readership working on IoT security, edge AI deployment, and explainable machine learning, and we thank you for considering it for publication.

Sincerely,

Subarmaniam Kannan (corresponding author) Sumendra Yogarayan